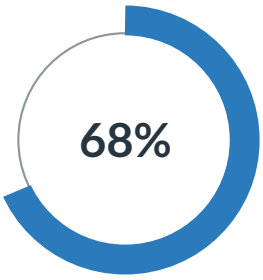
Results

Carter Feng

This assessment has been regraded. Questions 4, 6 have been affected.



34
Out of 50 points

40:11
Time for this attempt

Your Answers:

1 0 / 0 points Fill in the Blank

At Seven Lakes High School, we believe that character matters and that academic integrity reflects who we are as learners and citizens. By signing this pledge, I commit to upholding the values of honesty, responsibility, and respect in all academic and personal endeavors.

My Academic Integrity Pledge:

- **I will complete all academic work honestly, without giving or receiving unauthorized assistance.**
- **I will not copy, plagiarize, or submit someone else's work as my own.**
- **I will not use unauthorized materials (notes, devices, online resources) during assessments.**
- **I will respect the integrity of others' work and will not aid others in academic dishonesty.**
- **I will not share or receive academic work, ideas, or answers electronically or in-person when individual work is expected or collaboration is not permitted.**
- **I will report any knowledge of cheating or violations of academic integrity to a trusted adult.**

I understand that academic dishonesty harms my character, undermines trust in our community, and may result in consequences outlined in the school's academic integrity policy.

Print your name to acknowledge the oath. Not printing your name will result in a zero for the assessment.

✓ Carter Feng

2 5 / 5 points Fill in the Blank

Create a Binary Search Tree with the following values: 19, 5, 27, 44, 34, 40, 14, 38, 49, 17, 25, 22, 26, 50, 2.

What is the post order traversal of the tree?

✓ 2 ✓ 17 ✓ 14 ✓ 5 ✓ 22 ✓ 26

✓ 25 ✓ 38 ✓ 40 ✓ 34 ✓ 50 ✓ 49

✓ 44 ✓ 27 ✓ 19

3 5 / 5 points Fill in the Blank

Create a Binary Search Tree with the following values: 19, 5, 27, 44, 34, 40, 14, 38, 49, 17, 25, 22, 26, 50, 2.

How many nodes in the tree are degree 1?

✓ 4

This question has been regraded.

4 Previous score 5 / 5 points Regrade score 5 / 5 points Fill in the Blank

Create a Binary Search Tree with the following values: 24, 13, 45, 23, 9, 21, 36, 34, 8, 47, 16, 39, 22, 31, 4.

Remove the following values: 9, 24, 47, 21, 36.

What are the nodes on level 3 of the tree?

✓ 4 ✓ null ✓ 22 ✓ null ✓ 34 ✓ null ✓ null ✓ null

5 5 / 5 points Fill in the Blank

Create a Binary Search Tree with the following values: 19, 5, 27, 44, 34, 40, 14, 38, 49, 17, 25, 22, 26, 50, 2.

What is the number of nulls on level 4 of the tree?

✓ 14

This question has been regraded.

6 Previous score 0 / 5 points Regrade score 1 / 5 points Matching

Create a red-black tree using the following values:36, 4, 22, 27, 4

What are the following nodes values when the 2nd outside-outside ro

Red Parent Node	✗ 7	Correct Answer: **31**
Black Sibling Node	✓ null	
Red Child (or X) Node	✗ 17	Correct Answer: **34**
Black Grandparent Node	✗ 4	Correct Answer: **27**
What type of rotation is it?	✗ left - left	Correct Answer: **right-right**

7 5 / 5 points Fill in the Blank

Create a red-black tree using the following values:36, 4, 22, 27, 46, 7, 39, 17, 31, 3, 34, 1, 14, 8, 5.
What is on level 1 of the red-black tree?

Enter null when nothing is present.
Only enter value, not color.
Enter values in row major order.

✓ 7 ✓ 36

8 0 / 5 points Essay

Write the leftRightRotation method for a RedBlackTree class.

```
private RedBlackNode leftRightRotation(RedBlackNode g, RedBlackNode
{
```

```
private RedBlackNode leftRightRotation(RedBlackNode g, RedBlackNode p, RedBlackNode x) {
  p.right = rotateLeft(p.right);
  rotateLeft(p);
  return rotateRight(g);
}
private RedBlackNode rotateLeft(RedBlackNode p) {
  RedBlackNode r = p.right;
  p.right = r.left;
  if (r.left != null) {
    r.left.parent = p;
  }
  r.left = p;
  r.parent = p.parent;
  p.parent = r;
  return r;
}
private RedBlackNode rotateRight(RedBlackNode g) {
  RedBlackNode l = g.left;
  g.left = l.right;
  if (l.right != null) {
    l.right.parent = g;
  }
  l.right = g;
  l.parent = g.parent;
  g.parent = l;
  return l;
}
```

Incorrect

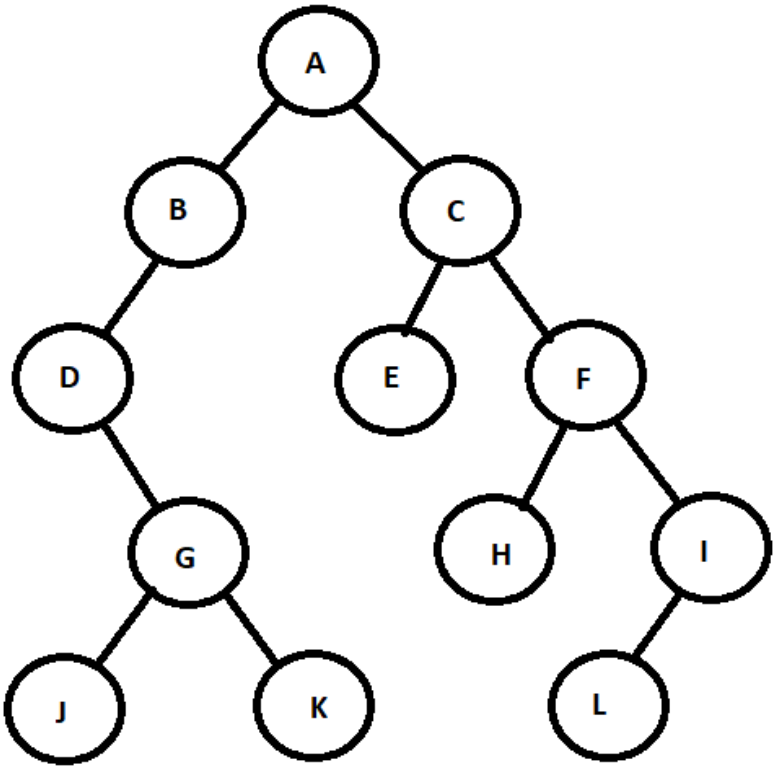
9 0 / 5 points Matching

Create a red-black tree using the following values:36, 4, 22, 27, 4

What are the following nodes values when the 4th outside-outside ro

Red Parent Node	✗ 3	Correct Answer: **14**
What type of rotation is it?	✗ right - right	Correct Answer: **left - left**
Red Child (or X) Node	✗ 1	Correct Answer: **8**
Black Grandparent Node	✗ 4	Correct Answer: **17**
Black Sibling Node	✓ null	

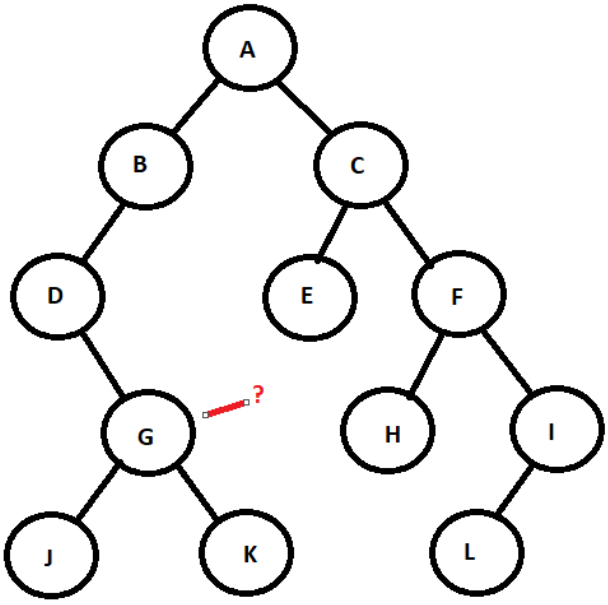
10 1 / 1 point Fill in the Blank



What is the diameter of this particular tree?

✓ 9

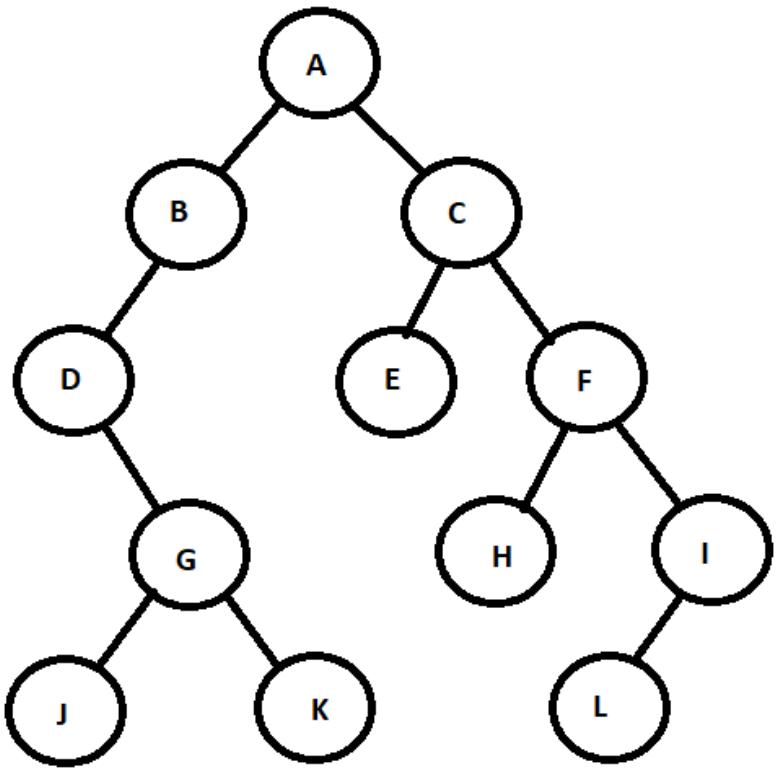
11 1 / 1 point Multiple Answer



Identify the marked item [?].

- [x] parent, internal node, inner node, branch node
- [ ] path
- [ ] leaf, outer node, terminal node
- [ ] left child
- [ ] root
- [ ] edge
- [x] right child
- [ ] sibling

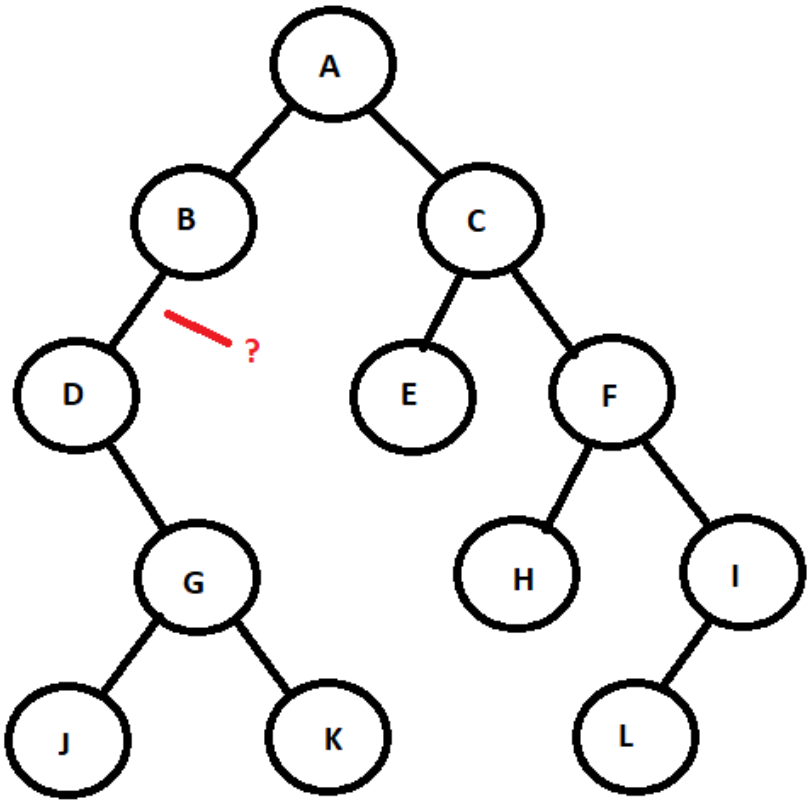
12 1 / 1 point Fill in the Blank



What is the level location of node J

4

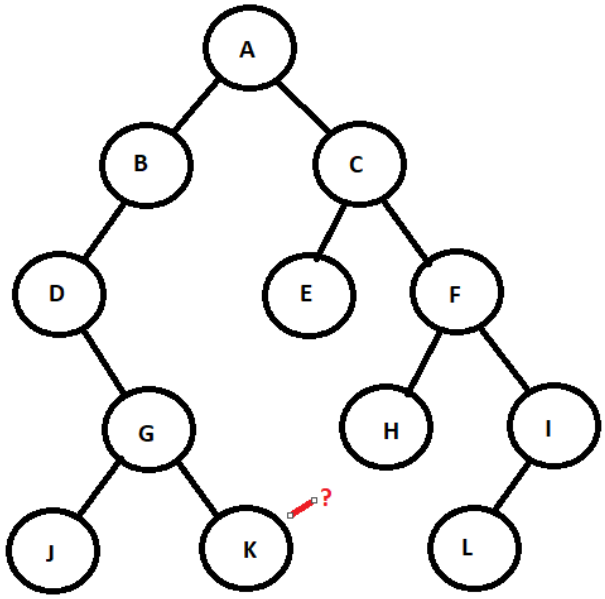
13 1 / 1 point Multiple Answer



Identify the marked item [?].

- [ ] sibling
- [ ] parent, internal node, inner node, branch node
- [ ] branch, subtree
- [ ] path
- [ ] root
- [ ] leaf, outer node, terminal node
- [ ] right child
- [x] edge
- [ ] left child

14 1 / 1 point Multiple Answer



Identify the marked item [?].

- [ ] root
- [x] leaf, outer node, terminal node
- [x] right child
- [ ] path
- [ ] edge
- [x] sibling
- [ ] parent, internal node, inner node, branch node
- [ ] left child

15 0 / 1 point Multiple Choice

What is the big O time constraint for adding a value to a binary search tree?

- ( ) O(N log N)

O(1)

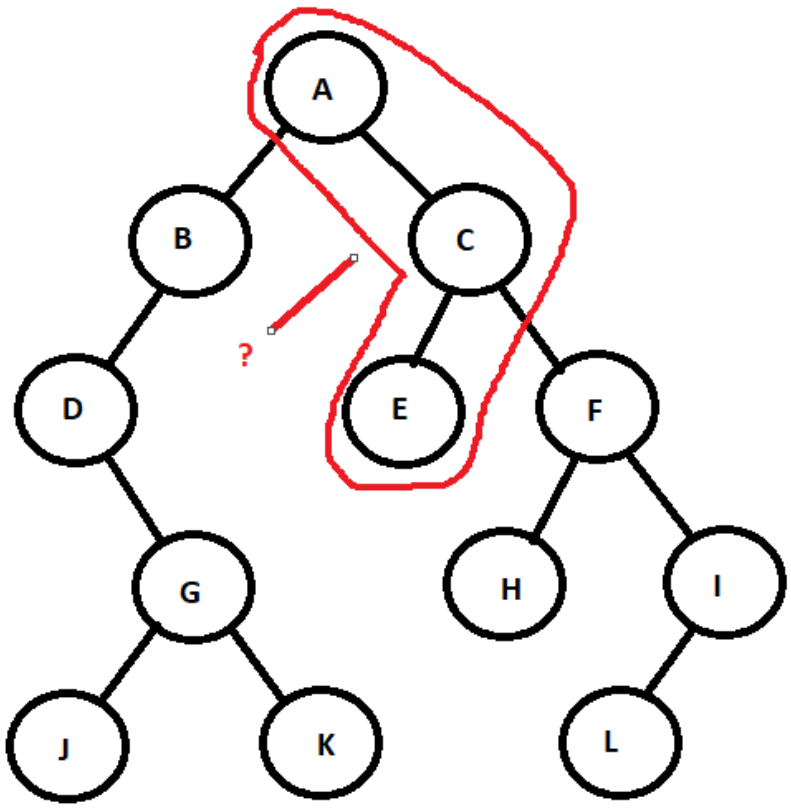
$O(N^2)$

✕ O(N)

Correct Answer: **O(log N)**

O(log N)

16 0 / 1 point Multiple Answer



Identify the marked item [?].

sibling

- [ ] edge
- [ ] leaf, outer node, terminal node
- [ ] parent, internal node, inner node, branch node
- [ ] left child
- [x] path

Missed Option - Incorrect

- [x] branch, subtree

Selected Answer - Incorrect

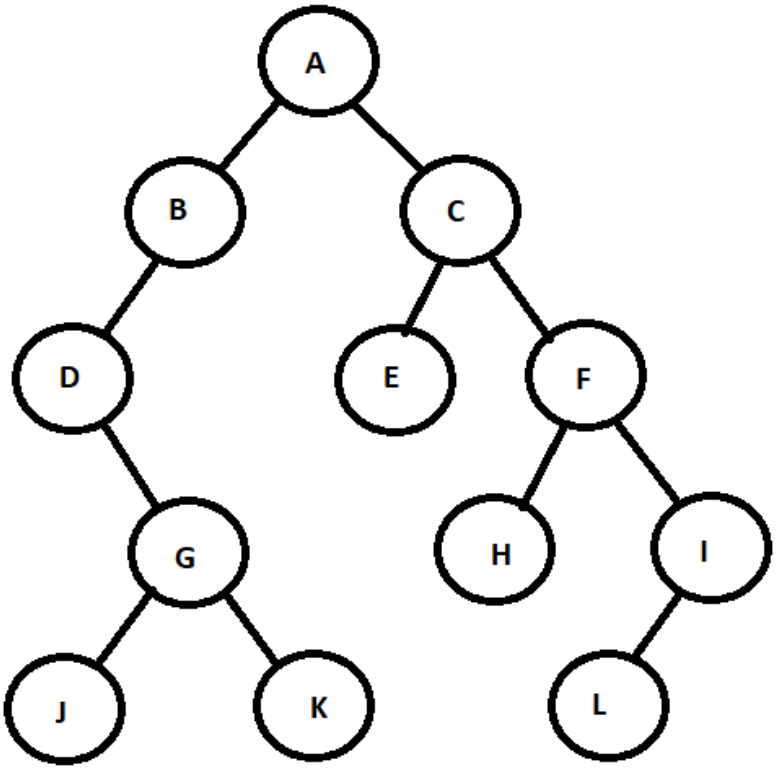
- [ ] root
- [ ] right child

17 1 / 1 point Multiple Choice

What is a full tree?

- ( ) A tree whose leaves are at the same depth.
- ( ) A tree whose levels are at maximum width, with the exception of the lowest level where all the nodes in that level are as far left as possible
- ( ) A tree that whose left subtree and right subtree heights differ by no more than 1 level
- (•) A tree whose only have nodes of degree 0 and degree 2

18 1 / 1 point Fill in the Blank



What is the level location of node E

✓ 2

19 1 / 1 point Multiple Choice

What is a balanced tree?

- A tree whose leaves are at the same depth.
- A tree whose levels are at maximum width, with the exception of the lowest level where all the nodes in that level are as far left as possible
- A tree whose only have nodes of degree 0 and degree 2
- ✓ A tree that whose left subtree and right subtree heights differ by no more than 1 level